Can we 'stop' global warming?

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August 9, 2026

Abstract

This report produces a simple mathematical model to investigate whether climate change can be 'stopped' by altering economic growth, technology, and implementing active carbon removal strategies. The model simplifies the Kaya Identity and combines the expression with a linear relationship between carbon emissions and temperature, as well as a model which represents the effects of temperature on sea levels. This report examines the mathematical and practical feasibility of three GICCE climate goals. Results indicate that improvements in technology and reduced economic growth can substantially slow warming, but stabilising temperatures requires net emissions to approach zero, and reducing temperatures requires sustained net-negative emissions. The most ambitious goal requires carbon removal to exceed 90 gigatonnes annually by the end of the century. All scenarios for achieving climate stabilisation are mathematically achievable within the model, however, few are likely to be practically feasible under current technological and economic conditions.

Contents

1 Introduction

Climate change is one of the most significant global environmental challenges, driven largely by anthropogenic greenhouse gas emissions from fossil fuel combustion, industrial processes and land-use changes [**ipcc_2023**]. The resulting emission of carbon dioxide (CO_2) into the atmosphere has caused a sustained rise in global mean temperature, leading to widespread environmental challenges such as ecosystem degradation, extreme weather events and sea-level rise. As part of the initiative of the Global Institute for Climate Change and the Environment (GICCE), this report seeks to improve general understanding of the Earth's climate and develop a simple mathematical model which reflects the goal of international governments to 'stop' climate change.

Human activities are the primary driver of increasing concentrations of CO_2 in the atmosphere [**gcb_2025**]. Fossil fuel combustion, land-use changes and other industrial processes release large quantities of carbon which exceed the Earth's natural ability to absorb it through carbon sinks [**gcb_2025**]. Emissions of CO_2 are determined by four key factors, being population, GDP per capita, energy intensity and carbon intensity [**deutch_2017**]. Although technological progress has reduced the carbon intensity of energy by approximately 0.7% per year over the past decade, these improvements have been outweighed by increasing global energy demand as a result of the growing population [**gcb_2025**]. Global anthropogenic CO_2 emissions have therefore continued to increase by 0.3% per year in the decade 2015–2024. Encouragingly, however, fossil fuel emissions have decreased in 35 economies during the decade 2015–2024 and emissions have slowed from 1.9% in the decade 2005–2014, demonstrating the success of partial decoupling of economic growth from emissions through technological changes.

Increased emissions of CO_2 into the atmosphere contribute substantially to higher concentrations of greenhouse gases which trap infrared radiation and thereby raise the global mean temperature. Global warming is roughly proportional to the cumulative sum of annual carbon emissions [**matthews_2009**], the constant of proportionality being the Carbon-Climate Response (CCR). This relationship forms the basis of the Global Carbon Budget 2025[**gcb_2025**] (GCB), which aims to reduce cumulative emissions to limit long-term warming. The GCB estimates that each additional emission of 180 $GtCO_2$ will lead to around 0.1°C of warming, leaving 170 $GtCO_2$ remaining from the beginning of 2026 for a 50% chance of limiting warming to 1.5°C, which is equivalent to roughly four years of emissions at the current rate [**gcb_2025**].

The rising global mean temperature has created significant environmental consequences, the particular focus of this report being sea-level rises. Sea-level rise continues even after temperature stabilisation due to the delayed response of oceans and melting ice sheets. The rate of sea level rise is approximately proportional to global temperature change over long time scales, so sea level rise can be modelled empirically from temperature [**rahmstorf_2007**]. If warming is limited to 1.5°C above the pre-industrial baseline, substantial long-term sea level rise is still expected, and is projected to rise approximately 2–3 metres over the next 2,000 years, although there is low confidence in the exactness of these projections [**gcb_2025**]. Reducing future emissions has become increasingly important to achieve temperature stabilisation and minimise long term environmental impacts.

2 Literature Review

The mathematical model produced in this report will quantify interactions between human activity and the climate system to assess future climate scenarios. The following literature will be considered according to how effectively each model captures the relationships required for this investigation and how appropriate their assumptions are for our simple model.

matthews_2009 (**matthews_2009**) related CO_2 emissions to global temperature. Using coupled climate-carbon cycle models, the authors found a linearly proportional relationship between global warming and cumulative carbon emissions from semi-empirical sources. The CCR is estimated to be approximately 1–2.1°C of warming per trillion tonnes of carbon emitted[matthews_2009]. This relationship is independent of time and atmospheric concentrations. Further, the model captures the processes of the carbon cycle, and offsets the effects of carbon uptake into the atmosphere on temperature against carbon sink saturation. This makes cumulative emissions highly useful for examining scenarios in which temperature is required to stabilise. However, **matthews_2009** calculated that there is substantial uncertainty on the model, being 0.4–1.5 $TtCO_2$. This is due to the various assumptions on which the relationship relies. For example, the authors assume that other greenhouse gases and non-greenhouse gases contribute negligibly to global warming. Further, estimates for historic land-use changes and aerosol forcing from limited available data is assumed to be accurate. However, uncertainty in this data would have great effects on the CCR value. The authors also note that their results are derived from idealised climate-carbon model scenarios over decades to centuries, meaning that extrapolation over longer time scales may not produce accurate results. The model produced in this report will only rely on **matthews_2009**'s linear relationship, and will estimate a constant of proportionality which is solely based on empirical data.

The Kaya Identity provides a complementary expression for determining annual CO_2 emissions. The identity decomposes annual CO_2 into a product of population, GDP per capita, energy intensity and carbon intensity. **deutch_2017** (**deutch_2017**) uses this expression to examine the relationship between economic growth and emissions, and demonstrates that improvements in energy and carbon intensity can weaken the relationship between economic growth and CO_2 emissions. This is important for the purposes of the current report as it allows a framework for proposing technological changes to produce a reduction in annual emissions. However, the Kaya Identity is not a causal model, so it cannot predict how the terms in the model will respond to technological development, government policy or changes in the markets. So, the reliability of the projections in that article are questionable. This report aims to adopt the relationship established by the Kaya Identity. However, it will assume that technological advancements have a direct impact on energy and carbon intensity. The GCB provides empirical critiques against the assumptions of the model. **gcb_2025** (**gcb_2025**) states that global emissions have not yet reached sustained decline despite energy and carbon intensity improvements, but also identifies evidence of partial decoupling. Our model must be closely examined against observed trends.

Population represents a fundamental driver of emissions due to rising demands for energy. **lee_2003** (**lee_2003**) summarised the demographic transition theory, which describes the growth changes of a population as a region develops. The paper is helpful to produce a logistics model for long-term demographic projections. However, the literature identifies several factors which would affect population trajectories and sheds important authority that a carrying capacity in a simple logistic model may not accurately represent the social, economic demographic mechanisms which would affect future population. Our logistic model is therefore used as an approximation for long-term behaviour and assumes that such behaviour is stable over long timescales.

Stabilising global temperature does not capture all risks associated with climate change. Sea-level rise continues long after temperatures stabilise due to delayed responses of ocean heat uptake and ice-sheet dynamics. **rahmstorf_2007** (**rahmstorf_2007**) proposed a semi-empirical linear relationship between sea-level changes and temperature. The model captures long-term sea level behaviour as opposed to an assortment of underlying physical processes. While the model performs well over longer timescales, it is less reliable over shorter periods as it does not account for fluctuations in physical processes. Nonetheless, it provides an appropriate framework for assessing whether sea-levels can eventually stabilise under changing climate scenarios.

The literature provides key relationships required for the model in this report, which will be applied to three scenarios by the GICCE. The first is stabilisation of global temperatures to 4°C above pre-industrial levels by 2100. The second is stabilisation at 2°C by 2050, and the third is limiting warming to no higher than 1.5°C while plateauing sea-level rise by 2100. The model is subsequently used to assess the various changes and advancements required to achieve each scenario and evaluate the implications for the environment and humanity.

3 Modelling Methodology

3.1 Main Models

To model temperature change, we modified the Transient Climate Response to Cumulative Carbon Emissions (TCRE) relationship[matthews_2009]. **matthews_2009** established a linear relationship between Global Surface Temperature anomaly and Cumulative Carbon Emissions, both since pre industrial levels, which we model as

$$\Delta T(t) \propto \int_{t_0}^{t} E(y)dy, \tag{1}$$

where:

- t is time in years;
- t_0 is the year we start measuring from;
- $\Delta T(t)$ is the temperature change at time t since t_0; and
- $E(t)$ is the amount of CO_2 produced by humans in year t measured in tonnes.

Note that emissions and CO_2 emissions are used interchangeably, as CO_2 accounts for a majority of the global heating due to greenhouse gas emissions, and a majority of greenhouse gases emitted [**noaa_2025**].

We modify this by introducing a reduction term that subtracts from the emissions, representing human/technological efforts to directly remove carbon from the atmosphere. This leads to a new equation, which is the primary model for the paper

$$\Delta T(t) \propto \int_{t_0}^{t} (E(y) - R(y))dy, \tag{2}$$

where R is a function representing annual carbon reduction efforts. To model anthropogenic carbon emissions E, we used the Kaya Identity[deutch_2017], which states

$$E(t) \propto \text{Population} \times \frac{\text{GDP}}{\text{Population}} \times \frac{\text{Energy used}}{\text{GDP}} \times \frac{\text{CO}_2\text{ produced}}{\text{Energy used}}.$$

In other words, CO_2 emissions is the product of population, GDP per capita, energy intensity, and carbon intensity. To simplify our model, we capture the product of the latter two as technological inefficiency D in terms of GDP per tonne of CO_2. This gives a simplified Kaya Identity

$$E \propto PYD, \tag{3}$$

where:

- P is population;
- Y denotes global average GDP per capita; and
- D is the technological inefficiency term.

3.2 Sub-Models

Next, we look at these terms in more detail.

3.2.1 Population

Population is not an aspect of the model that will be altered as a parameter in the scenarios. The first iteration of modelling this was using a logistic equation, giving population as

$$P(t) = \frac{P_{\text{final}}}{1 + (\frac{P_{\text{final}}}{P_{\text{initial}}} - 1)\exp\left(-r_p(t - 1950)\right)},$$

where:

- 1950 is the start of the modelling period;
- P_{initial} is the population in 1950;
- P_{final} is a carrying capacity initially set to 10 billion, based on UN estimates[un_population_2024]; and
- r_p is the rate of population growth, which has been fitted[our_code] to ≈ 0.031 in our model.

However, a weakness of logistic equations in modelling human population was that it did not capture realistic projections of human populations, which are projected to decline by the end of the century[un_population_2024], while the logistic equation tends asymptotically to the carrying capacity.

To better model the projections of falling populations in the future, we added a second logistic equation that subtracts from the original one. This second term is referred to as the population decline term, giving us a final population model

$$P(t) = \frac{P_{\text{final}}}{1 + (\frac{P_{\text{final}}}{P_{\text{initial}}} - 1)\exp\left(-r_p(t - 1950)\right)} - \frac{P_{\text{dec\_final}}}{1 + (\frac{P_{\text{dec\_final}}}{P_{\text{dec\_init}}} - 1)\exp\left(-r_{\text{dec\_p}}(t - 2050)\right)}, \tag{4}$$

where:

- $P_{\text{dec\_init}}$ is set to 250 million;
- $P_{\text{dec\_fin}}$ is the amount population will decline, set to 2 billion for our model;
- $r_{\text{dec\_p}}$ is set to 0.04; and
- P_{final} is updated to 12 billion, letting the asymptotic population remain 10 billion.

3.2.2 Economy

The economic term Y, in terms of GDP per capita, is one of the parameters modelled that will be adapted to meet scenarios. It is modelled simply as an exponential, given as

$$Y(t) = 1000 + A_y e^{r_y t}, \tag{5}$$

where:

- A_y is an adjustment factor to fit historical data;
- r_y represents the rate of growth of GDP / capita; and
- \$1000 is the historical baseline GDP, which was a quirk of the data used[owid_gdppc].

The terms A_y and r_y were fitted to historical data**our_code**.

In the modelling scenarios, Y is treated as a piecewise continuous function where, starting from 2027, the function's growth rate reduces to a fraction of the original growth rate.

The choice of exponential is appropriate for the period of time the model is relevant for.

While GDP is effectively related to population, as a larger population implies a rise in GDP, GDP per capita represents income per person, which is influenced more greatly by technological and productivity improvements than population.

Technological and productivity gains are in turn, derived from a sizable working age population that continually innovate and improve [**lindh_1999**]. A sizable working age population will remain so long as there is a young workforce present in the world, and this is the case for at least until 2100, even with the diminishing population towards the end of the century [**un_population_2024**].

3.2.3 Technological Inefficiency

This term D broadly represents the emitted CO_2 per unit of GDP. As research continues in the future, improvements in technology should result in lower carbon and energy intensities, lowering this value. It is modelled as the exponential

$$D(t) = A_d e^{r_d t}, \tag{6}$$

where:

- A_d is an adjustment factor; and
- r_d represents the rate of decreased inefficiency each year.

Both of these terms were fitted to historical data**our_code**.

The technology term is also used as a parameter in the scenarios, where the rate of growth of technology is altered starting from the year of action (2027). For example, in scenario 1, we model technological improvements to be 1.5 times faster, directly improving energy intensity and carbon intensity 1.5x more than original projections.

3.3 Reduction Term

In Equation 2, a reduction term was introduced. This represents another technological aspect to the model, representing human efforts to remove carbon from the atmosphere. Initially, natural carbon sinks were considered to be a part of the reduction term, but the TCRE relationship**matthews_2009** implicitly accounts for natural sink behaviour in its model, specifically using cumulative human emissions.

Hence, the reduction term must also denote human efforts to remove carbon from the atmosphere. Examples of initiatives include engineering solutions such as Direct Air Capture, and nature based solutions such as reforestation and regenerative agriculture.

In our scenario modelling, the reduction term is the primary driver to offsetting the carbon emissions. Constraining economic growth and improving carbon and energy intensities only reduce the amount of emissions; they do not eliminate it. Achieving net zero targets in the model requires significant carbon removal from the atmosphere, and hence the reduction term is used predominantly in the model.

When modelling scenarios, the term will use a rising function, representing increasing capacity of technology to remove carbon from the atmosphere and construction effort as time moves forward. This will be adjusted to the scenarios to fit each target.

3.4 Sea Level

Scenario 3 requires sea levels to stop rising by 2100. For this, **rahmstorf_2007**'s semi empirical approach to projecting sea level rise was used. Here, **rahmstorf_2007**'s model approximated the rate of change of sea levels to be proportional to the temperature anomaly from pre industrial levels to predict long term trends, which solves our purpose. **rahmstorf_2007**'s model stated

$$\frac{dH}{dt} = \alpha(T - T_0), \tag{7}$$

where:

- H represents sea level;
- T is the global surface temperature;
- T_0 is the pre-industrial baseline temperature; and
- α is the proportionality constant, which was calculated[rahmstorf_2007] to be $3.4\frac{mm}{year}$ per degree Celsius.

This model simplifies by not accounting for the lag between temperature and sea levels, mainly considering long-term trends and averages [**rahmstorf_2007**].

4 Results

We have been given a set of scenarios to model, each with specific temperature targets and dates. So we simulate efforts to actively remove carbon from the atmosphere from 2027. In addition to the active carbon removal, we also adjust the growth rate of Y and decay rate of D after that year to find combinations of economy growth change, technological improvements, and active removal that satisfy the scenarios.

4.1 Scenario 1

For a maximum target of 4°C above pre-industrial levels by 2100, we reduced the economy growth rate r_y to 70% of its original value and increased the technology improvement rate r_d by 1.5 times.

Then we set our active removal R to linearly ramp up to 3 $GtCO_2$ per year by 2077 and continue, reaching and remaining at 5 $GtCO_2$ per year.

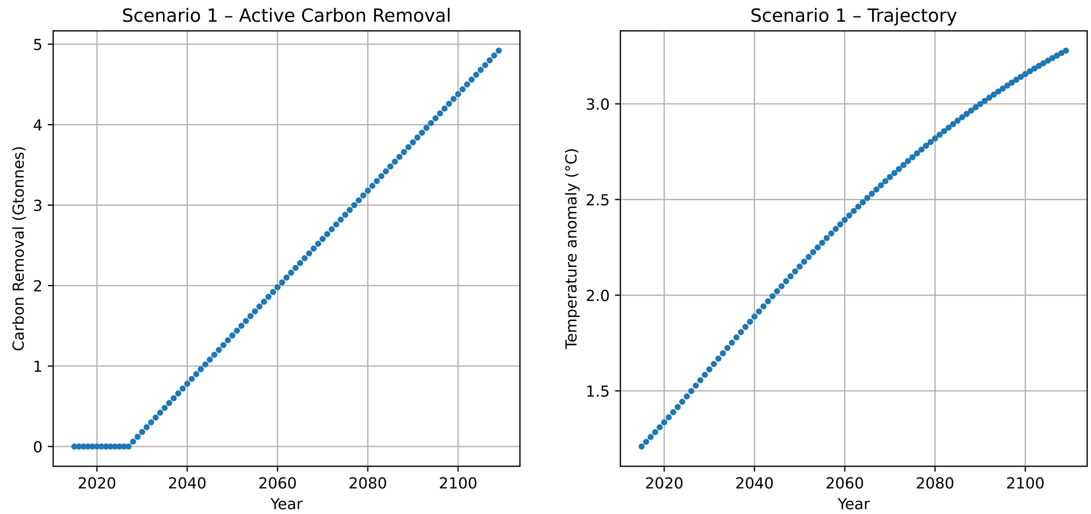

Figure 1: Scenario 1 active CO_2 removal and resulting temperature projections

So in this model, the temperature anomaly remains under 4°C until the year 2250, plateauing from there (see Appendix). Reducing the effectiveness of carbon removal by 90% causes the temperature to increase by 3.17°C in 2100, plateauing above 4.1°C after 2250.

4.2 Scenario 2

For a maximum target of 2°C above pre-industrial levels by 2050, we reduced the economy growth rate r_y to 60% of its original value and increased the technology improvement rate r_d by 2 times.

Then we set our active removal R to be a growing portion of the annual emissions. This ramps up to nearly 100% by 2050, making it plateau afterward.

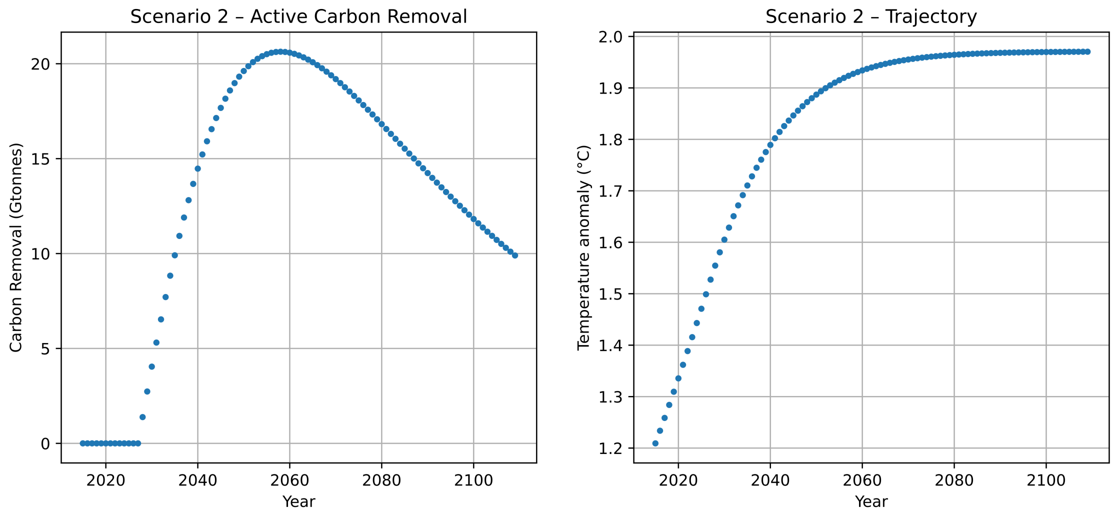

Figure 2: Scenario 2 active CO_2 removal and resulting temperature projections

So in this model, the temperature change remains under 1.9°C in 2050, and remains under 2°C. If we consider a situation where removal is only 90% of the target, the model reaches 2°C by 2070, and reaches about 2.1°C.

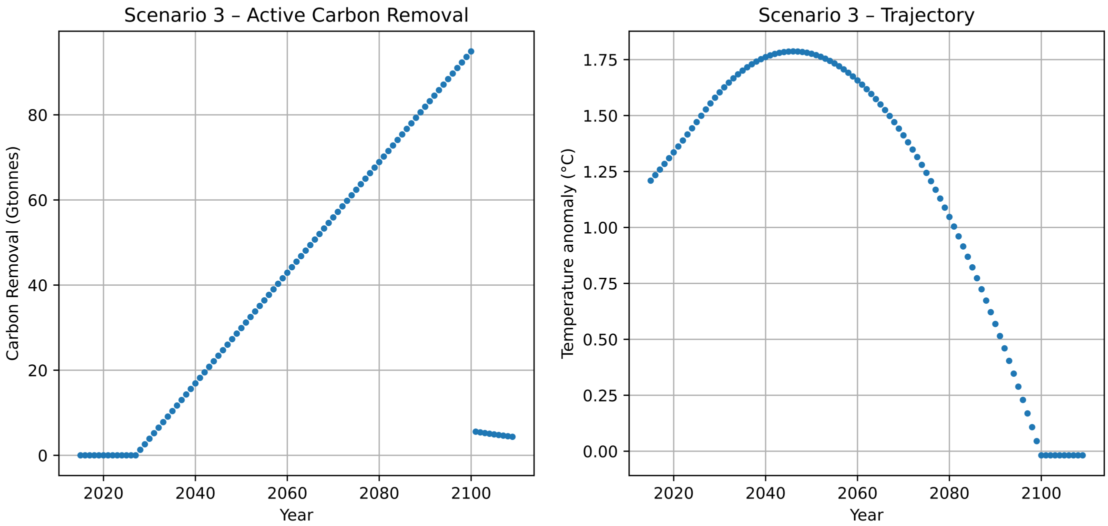

Figure 3: Scenario 3 active CO_2 removal and resulting temperature projections

4.3 Scenario 3

This scenario requires a maximum temperature anomaly of 1.5°C and for sea levels to stop rising by 2100. Unfortunately, our temperature model in **??** indicates that we have already surpassed the 1.5°C target as of 2025. So we only consider the sea level requirement.

From **??**, we must have the temperature change reach 0°C by 2100 for this to occur. So we model more aggressively, first reducing the economy growth rate r_y to 50% of its original value and increasing the technology improvement rate r_d by 2.5 times. For the active carbon removal, we follow a linear ramp, reaching 94.9 gigatonnes of CO_2 by 2100, then lowering efforts to match global emissions in order to keep temperature stable.

This results in a peak above 1.75°C before 2050, which drops to 0°C before 2100. Using the modelled temperature from **??**, we can determine the expected sea levels. So from **??**, we approximate the sea level increase since 2015

$$H(t) = \alpha \int_{2015}^{t} (T(y) - T_0)dy,$$

with results shown in **??**. In this model, the sea levels stop rising at 381 mm higher than in 2015.

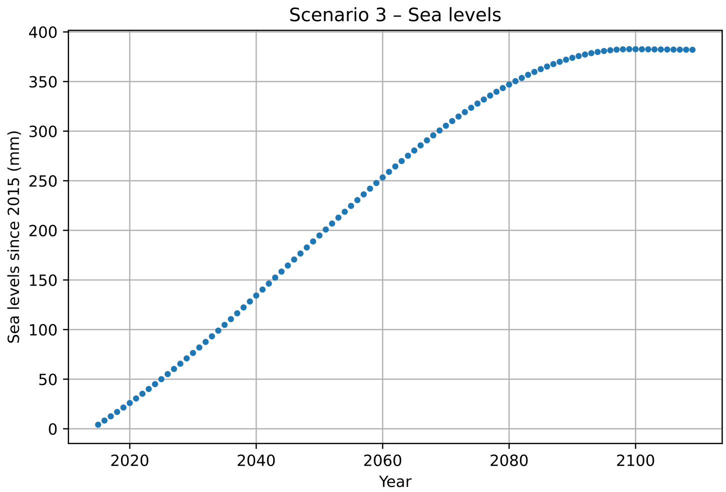

Figure 4: Scenario 3 model sea levels since 2015

If the removal efforts were only 90% as effective in this model, the linearly increasing effort to remove CO_2 would need to continue for four additional years in order to bring temperatures back to pre-industrial levels. This would also cause the projected sea level increase to 404 mm, about 23 mm higher than the original prediction.
An additional factor not considered in this model is the temperature lag for the ocean. The effects of temperature change may be reflected decades later in the ocean. Since this model would bring the temperature down before the ocean levels finished rising, the true peak sea level could be much lower than what is predicted here.

5 Discussion

Woah, is there any discussion to be had?

6 Conclusion

The model that we have created suggests that climate change can be stabilised. However more ambitious targets require much stronger intervention whereby stabilising temperature requires net emissions to approach net zero and reducing temperature requires sustained net-negative emissions. Scenario 1 is the most realistically feasible, and Scenario 2 requires more advancements in technology and carbon remove methods in order to happen. But Scenario 3, while it is mathematically possible, is not feasible in the real world due to its extreme carbon-removal requirements.

Our simplified model shows that technology and carbon removal must be complemented with rapid emissions reductions as opposed to outright replacing it. These targets are mathematically possible, but under current trends, they are not yet practically feasible.

7 Bibliography

A Appendix

A.1 Figures

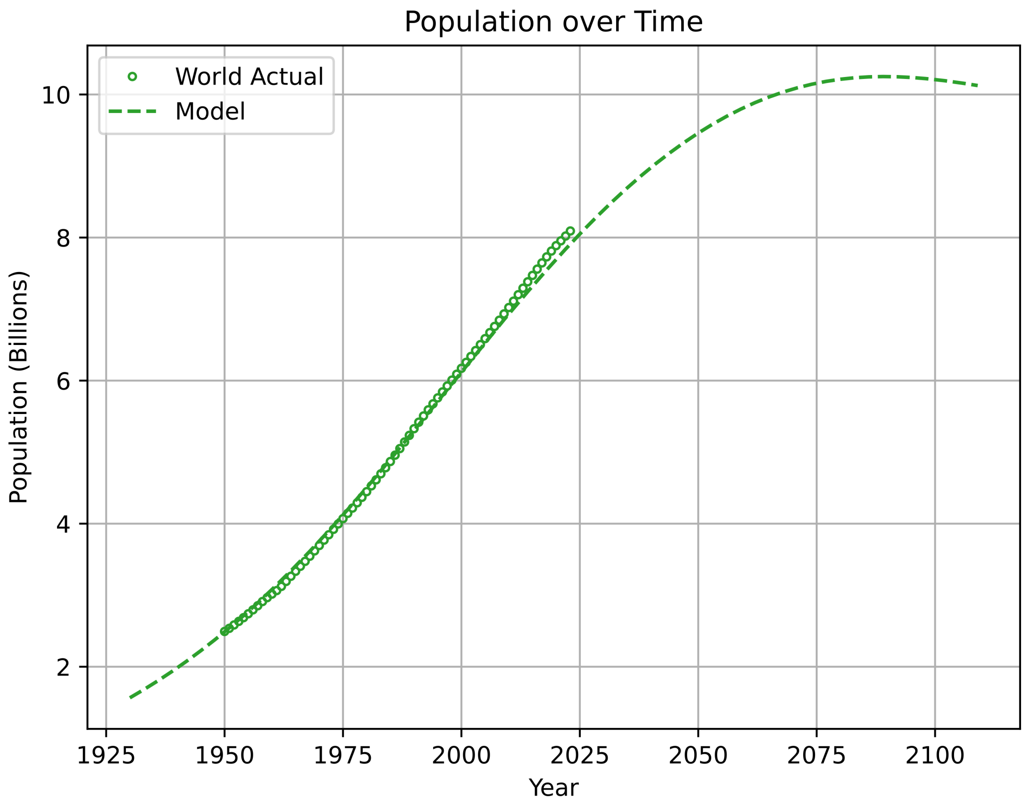

Figure 5: Population model compared to real world population, dataset from Our World in Data for years 1950–2023. mean percentage error = 0.0987%, percentage error stdev = 1.478%

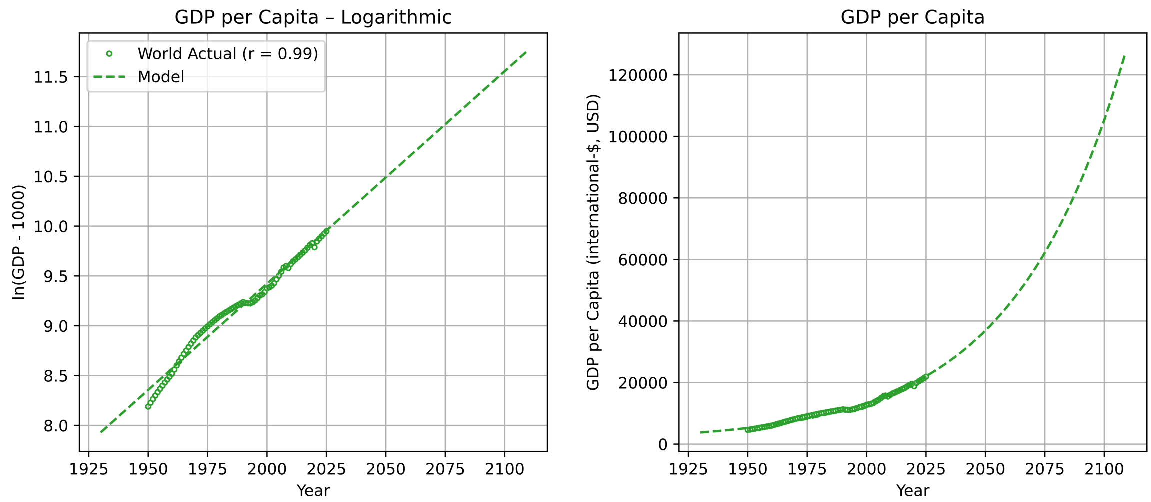

Figure 6: GDP per capita model compared to real world historical GDP, dataset from Our World in Data for years 1950–2025. mean percentage error = 0.1199%, percentage error stdev = 5.93%

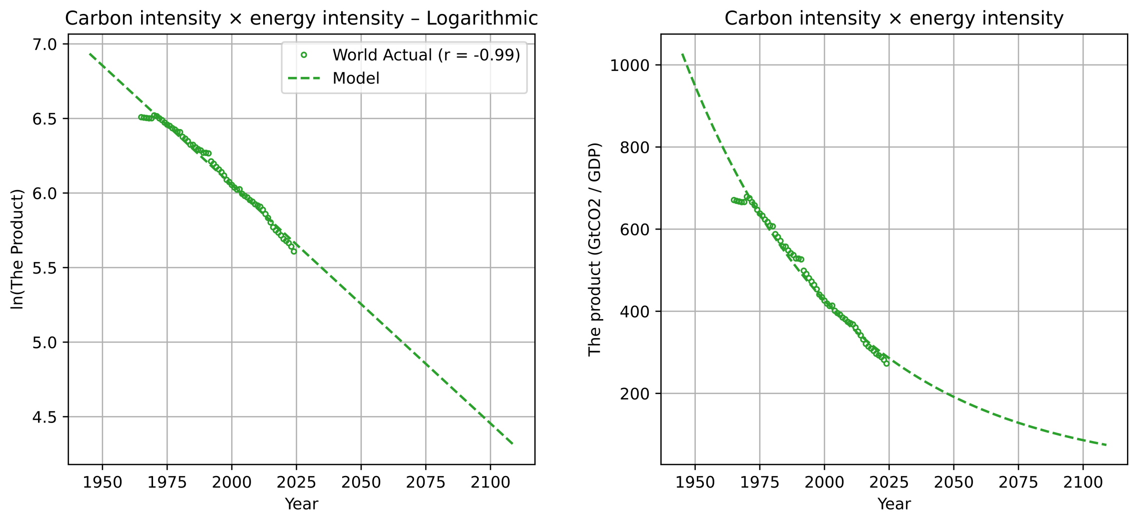

Figure 7: Technological inefficiency model compared to real world historical carbon intensity $\times$ energy intensity, dataset from Our World in Data for years 1965–2024. mean percentage error = 0.061%, percentage error stdev = 3.53%

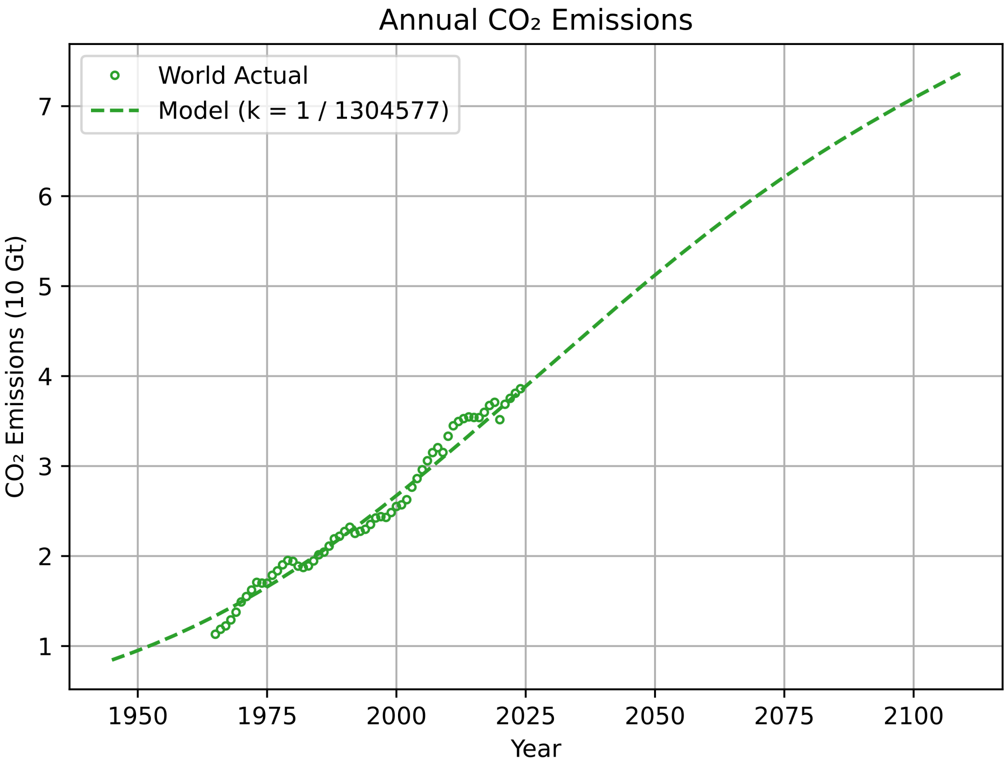

Figure 8: CO_2 model compared to real world historical CO_2 emissions, dataset from Our World in Data for years 1965–2024. mean percentage error = -1.935%, percentage error stdev = 5.5359%

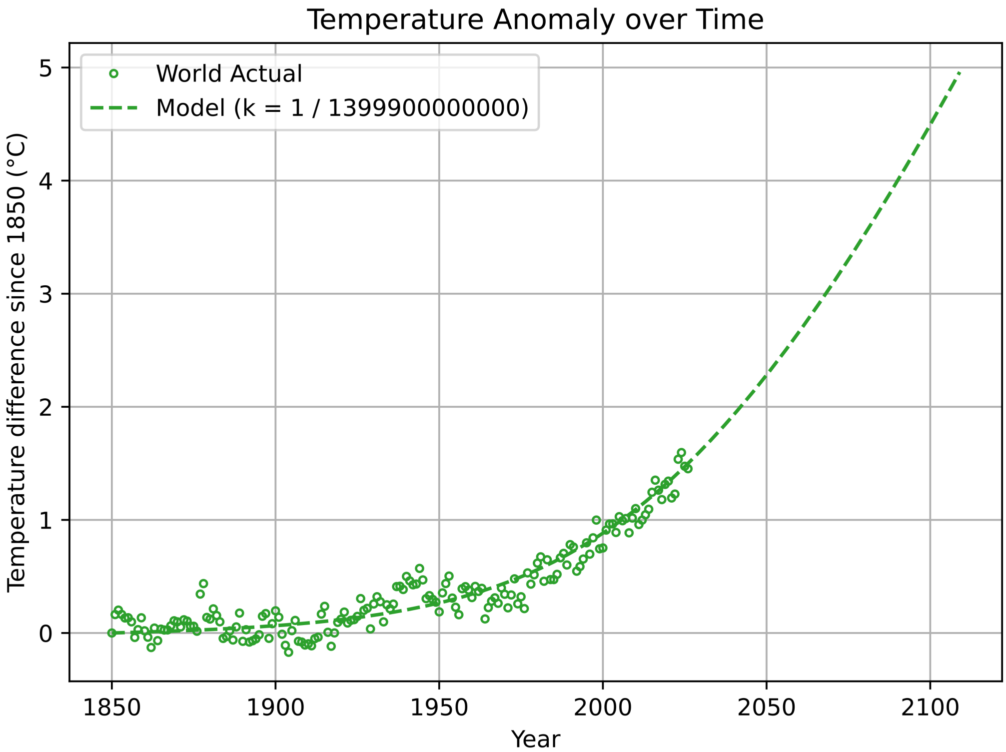

Figure 9: Temperature model compared to real world historical temperature anomaly, dataset from Our World in Data for years 1850–2026. mean error = <0.0003°C, error stdev = 0.1259°C

A.2 Our Code

We have included the code used to model in our bibliography **[our_code] our_code** (**our_code**).